

TEACHER ANSWER KEY

Peppered Moth: Natural Selection Explained | Year 9 KS3

Suggested answers are indicative. Credit scientifically correct alternatives supported by the video.

1. The peppered moth.
2. *Biston betularia*.
3. A pale, speckled form and a dark (melanic) form.
4. Colouring or patterns that help an organism blend into its surroundings and avoid detection.
5. Birds (accept another predator if it is identified in the video).
6. Pale lichens.
7. Their colouring matched the pale, lichen-covered bark, so birds were less likely to see and eat them.
8. Soot and pollution darkened the bark; industrial pollution also damaged or killed lichens.
9. Much of it died or disappeared because of air pollution.
10. Dark moths were better camouflaged against soot-darkened bark, so predators were less likely to see them.
11. The pale moths contrasted with the dark bark, making them easier for birds to spot and eat.
12. Dark moths.
13. More dark moths survived to reproduce. They passed inherited alleles for dark colour to offspring, so the proportion of dark moths increased over generations.
14. Predation by birds, acting together with the colour of the background/environment.
15. Individuals do not change an inherited colour because they need to. Variation already existed, and natural selection changed the proportions of colour forms across generations.
16. The bark became paler as soot decreased and pale lichens returned.
17. Pale moths became more common because their camouflage improved, so more survived and reproduced.
18. Dark moths will probably become more common because they are better camouflaged, experience less predation and leave more offspring. Accept any well-supported prediction.
19. Variation -> environmental change -> differential survival -> reproduction.
20. Answers should include: inherited colour variation existed; environmental change altered camouflage; better-camouflaged moths were less likely to be eaten; survivors reproduced and passed on their alleles; the population changed over generations.

Total: 39 marks